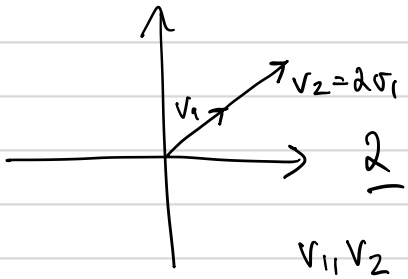
Lecture 8

Today: Linear independence.

Vectors $v_1, v_2, \dots, v_k$ are <u>independent</u> if no combination gives the zero vector (execpt the zero comb.):

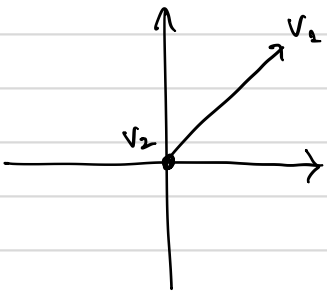
$$c_1 v_1 + c_2 v_2 + \dots + c_k v_k \neq 0$$

unless all $c_i = 0$



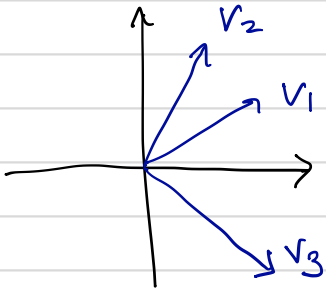
$$\underline{2}\, v_1 + \underline{(-1)}\, v_2 = 0$$

v_1, v_2 <u>dependent</u>.



$$\underline{0}\, v_1 + \underline{1}\, v_2 = 0.$$

dependent.



dependent.

$$\begin{bmatrix} 2 & 1 & 2.5 \\ 1 & 2 & -2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Fact: A with m rows, n columns.
If $m < n$, then $Ax = 0$ has a <u>nonzero</u> solution.
$\downarrow$
more unknowns than equations.

Reason: There will be free variables.

$v_1, v_2, \dots, v_n$ vectors in $\mathbb{R}^m$

$$A = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{bmatrix} \Big\} m \text{ rows.}$$

Elimination: $A \longrightarrow U$.

Prop: The following are equivalent:

1) $v_1, v_2, \dots, v_n$ are independent.
2) $Ax = b$ has <u>at most one</u> solution for all b. $\hookrightarrow$ 0 or 1
3) $N(A) = 0$
4) U has one pivot in every column ($\Rightarrow$ no free variables)

Prop: The following are equivalent:

1) $v_1, v_2, \dots, v_n$ span $\mathbb{R}^m$
2) $Ax = b$ has <u>at least one</u> solution for all b. $\hookrightarrow$ 1 or ∞
3) $C(A) = \mathbb{R}^m$
4) U has one pivot in every row. ($\Rightarrow$ no row of zeros)

No 5 vectors in $\mathbb{R}^4$ are independent
No 3 vectors in $\mathbb{R}^4$ span $\mathbb{R}^4$.

$$A = \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix} \xrightarrow{\text{Elim}} U = \begin{bmatrix} * & & \\ 0 & * & * \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

<u>Theorem</u>: $(m = n)$

$v_1, v_2, \ldots, v_n$ vectors in $\mathbb{R}^n$

$$A = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{bmatrix}$$

The following are equivalent:

1) $v_1, \ldots, v_n$ span $\mathbb{R}^n$
2) $v_1, \ldots, v_n$ are indepent
3) $Ax = b$ has exactly one solution for all b (A is <u>nonsingular</u>).

<u>Proof</u>: Do elimination: $A \longrightarrow U$.

① $\Longleftrightarrow$ 1 pivot in each row of U

$\Updownarrow$

n pivots

$\Updownarrow$

② $\Longleftrightarrow$ 1 pivot in each column of U

Vectors $v_1, v_2, \ldots, v_d$ form a <u>basis</u> of a vector space if

1) they span the space, and
2) they are independent.

The <u>dimension</u> of the space is the number of elements in any basis.